

Solution :

To find D :

$$D = \epsilon E = \epsilon_0 \epsilon_r E = 2.55 \epsilon_0 \times 10 \times 10^3 \text{ (C / m}^2\text{)} = 0.225 \mu\text{C / m}^2 \quad \dots\text{Ans.}$$

To find P :

$$D = \epsilon_0 E + P$$

$$P = D - \epsilon_0 E = 0.225 \times 10^{-6} - \frac{10^{-9}}{36\pi} \times 10 \times 10^3 = 0.137 \mu\text{C / m}^2 \quad \dots\text{Ans.}$$

To determine the surface charge density of the free charges :

(iii) We know that free surface charge density = $D = 0.255 \mu\text{C / m}^2$...Ans.

To find surface charge density of the polarization charge :

(iv) We have surface charge density of the polarization charge as :

$$\rho_{sp} = P = 0.137 \mu\text{C / m}^2 \quad \dots\text{Ans.}$$

To find potential difference :

(v) Potential difference, $V = Ed = 10 \times 10^3 \times 1.5 \times 10^{-3} = 15 \text{ Volts.}$...Ans.

5.6 Boundary Conditions :

MU - May 2010

In the previous sections, we have not come across problems involving more than one medium. In practice these type of situations occurs in case of reflections of waves at an air-dielectric interface, multiple dielectric capacitor and guiding of waves in metallic waveguide.

In case of different media, when field vector crosses the boundary surface, it is characterized by change in its magnitude and direction. This is true for electric as well as magnetic fields.

To solve the problem involving a boundary surface between different media, we should know the conditions satisfied by the field components at the boundary. These are known as boundary conditions.

These boundary conditions relates the field components at a point adjacent to and on one side of the boundary to the field components at corresponding point adjacent to and on the other side of the boundary.

Usually field components specified above are the tangential and normal components of the field vector.

How to handle the problems on boundary conditions ?

When the field is making angle α with the normal it can be resolved into two parts :

i) Tangential component to the interface (E_t)

ii) Normal component to the interface ($\bar{E}_n$)

where t and n subscripts indicate the tangential and normal respectively.

Knowing tangential and normal components we can obtain the total field.

For example, let $\bar{E}_1$ making angle α_1 with the normal is given and our problem is to determine $\bar{E}_2$. Refer Fig. 5.6.1(a). The approach is as follows :

- The given field $\bar{E}_1$ can be resolved into

- i) tangential component denoted by $\bar{E}_{t1}$

- ii) normal component denoted by $\bar{E}_{n1}$.

as shown in Fig. 5.6.1(b). Thus we know $\bar{E}_{t1}$ and $\bar{E}_{n1}$.

- Now $\bar{E}_2$ can be determined if we have $\bar{E}_{t2}$ and $\bar{E}_{n2}$. By having relation between tangential components, then knowing $\bar{E}_{t1}$ we can determine $\bar{E}_{t2}$.

- Similarly if we have relation between normal components then knowing $\bar{E}_{n1}$ we can go for $\bar{E}_{n2}$.

- Thus knowing $\bar{E}_{t2}$ and $\bar{E}_{n2}$ we obtain $\bar{E}_2$ as

$$\bar{E}_2 = \bar{E}_{t2} + \bar{E}_{n2}$$

The relationship between tangential and normal components discussed above is obtained using boundary conditions.

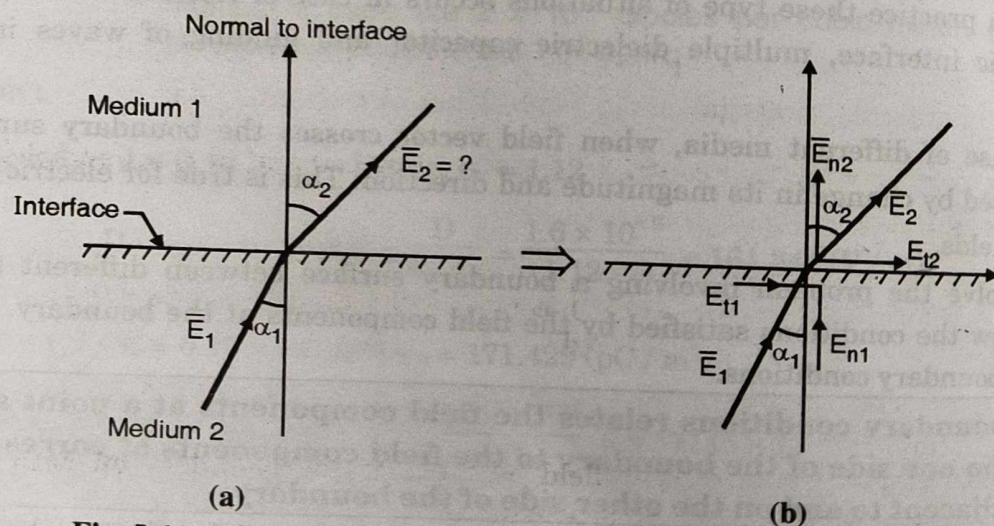


Fig. 5.6.1 : The problem given, (b) Approach to solve the problem

The boundary conditions changes when media changes. So we determine boundary conditions for

- i) dielectric - dielectric interface
- ii) dielectric - conductor interface

separately. We shall determine boundary conditions for electrostatics and magnetostatics separately. Similarly the boundary conditions for tangential and normal components is obtained separately.

5.7 Electrostatics Boundary Conditions :

Here first we shall consider interface between two dielectrics.

MU - May 2010

5.7.1 Dielectric-dielectric Interface :

Let us consider two semi infinite dielectric media separated by a plane boundary as shown in Fig. 5.7.1. Suppose $y = 0$ forms the boundary of medium 1 and 2 and these media have constants ϵ_1 and ϵ_2 respectively (σ_1 and σ_2 equal to zero). First we shall determine the boundary conditions for tangential components.

A) Tangential components :

Consider a rectangular path of length Δy and width Δx and placed half in each medium. Let the tangential components to the boundary be E_{t1} and E_{t2} , in medium 1 and 2 respectively. We know that

$$\oint \vec{E} \cdot d\vec{l} = 0$$

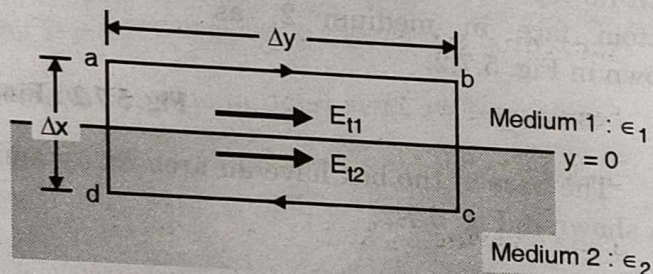


Fig. 5.7.1 : Showing imaginary rectangle, half in each medium

Applying this is to the rectangular path shown in Fig. 5.7.1, we get

$$\oint \vec{E} \cdot d\vec{l} = \int_{abcd} \vec{E} \cdot d\vec{l} = \int_{ab} \vec{E} \cdot d\vec{l} + \int_{bc} \vec{E} \cdot d\vec{l} + \int_{cd} \vec{E} \cdot d\vec{l} + \int_{da} \vec{E} \cdot d\vec{l} = 0$$

If we let side $bc = da = 0$, their contribution to the line integral of $\vec{E}$ around the path can be neglected. We have,

$$\oint \vec{E} \cdot d\vec{l} = \int_{ab} \vec{E} \cdot d\vec{l} + \int_{cd} \vec{E} \cdot d\vec{l} = 0 = E_{t1} \Delta x - E_{t2} \Delta x = 0$$

or

$$E_{t1} = E_{t2}$$

...(5.7.1)

i.e. tangential components of electric field adjacent to the boundary on both sides are equal and in same direction or in simple words.

Tangential components are continuous across the boundary of two dielectrics. This is the boundary condition for electric field.

$\vec{D}$ and $\vec{E}$ are related by $\vec{D} = \epsilon \vec{E}$, we have from Equation (5.7.1)

$$E_{t1} = E_{t2}$$

$$\text{or } \frac{D_{t1}}{\epsilon_1} = \frac{D_{t2}}{\epsilon_2}$$

$$\text{or } \frac{D_{t1}}{D_{t2}} = \frac{\epsilon_1}{\epsilon_2} \quad \dots(5.7.2)$$

Since media 1 and 2 are having different permittivity i.e. $\epsilon_1 \neq \epsilon_2$, then $D_{t1} \neq D_{t2}$ i.e. tangential components of $\vec{D}$ are not continuous across an interface.

B) Normal components :

In order to find relation between the normal components of the fields at a boundary, consider a small box half in each medium with its top face in medium 1 and bottom face in medium 2, as shown in Fig. 5.7.2.

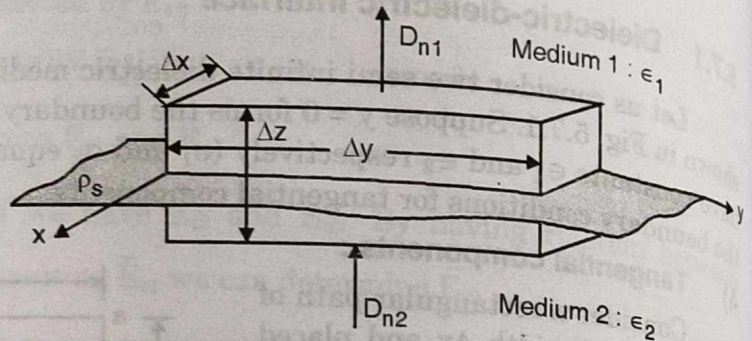


Fig. 5.7.2 : Finding relation between normal components

The base of the box have an area $\Delta x \Delta y$ and height Δz . Applying Gauss's law to the box shown in Fig. 5.7.2,

$$\oint_s \vec{D} \cdot d\vec{s} = \text{Charge enclosed}$$

When the boundary plane has surface charge density ρ_s , the area of the plane enclosed by the box is $\Delta x \Delta y$. Now we can determine charge enclosed,

$$\text{Charge enclosed} = \rho_s \Delta x \Delta y$$

$$\text{Therefore, } \oint_s \vec{D} \cdot d\vec{s} = \rho_s \Delta x \Delta y \quad \dots(i)$$

The left hand side is determined as :

$$\oint_s \vec{D} \cdot d\vec{s} = \oint_{\text{Top}} \vec{D} \cdot d\vec{s} + \int_{\text{Bottom}} + \int_{\text{left}} + \int_{\text{Right}} + \int_{\text{Front}} + \int_{\text{Back}} \dots(ii)$$

If we let height of the box vanishingly small, the contribution of the vertical faces will be zero and Equation (ii) reduce to

$$\oint_s \vec{D} \cdot d\vec{s} = \oint_{\text{Top}} \vec{D} \cdot d\vec{s} + \int_{\text{Bottom}} \vec{D} \cdot d\vec{s} \quad \dots(iii)$$

$$\text{For top surface : } d\vec{s} = \Delta x \Delta y \vec{a}_z; \quad \text{for bottom surface : } d\vec{s} = \Delta x \Delta y (-\vec{a}_z);$$

Thus,

$$\oint_{\text{Top}} \bar{D} \cdot d\bar{s} = D_{n1} \Delta x \Delta y$$

and

$$\oint_{\text{Bottom}} \bar{D} \cdot d\bar{s} = -D_{n2} \Delta x \Delta y$$

Putting these values of integrals in Equation (iii) and using Equation (i) we get,

$$D_{n1} \Delta x \Delta y - D_{n2} \Delta x \Delta y = \rho_s \Delta x \Delta y$$

or

$$D_{n1} - D_{n2} = \rho_s$$

...(5.7.3)

The above equation says that the normal component of $\bar{D}$ is discontinuous across the surface by the amount of surface charge density. When two dielectrics are in contact, the charge cannot be present at the boundary unless placed by mechanical means i.e. by rubbing, otherwise $\rho_s = 0$, giving

$$D_{n1} = D_{n2}$$

i.e. the normal component of the fixed density is continuous across the charge-free boundary between two dielectrics. Knowing $D = \epsilon E$, it follows that,

...(5.7.4)

$$\epsilon_1 E_{N1} = \epsilon_2 E_{N2}$$

Since $\epsilon_1 \neq \epsilon_2$, we get $\epsilon_{n1} \neq \epsilon_n$ i.e. normal $\bar{E}$ is discontinuous.

Boundary conditions for dielectric-dielectric interface

$$E_{t1} = E_{t2}$$

$$D_{n1} - D_{n2} = \rho_s$$

or

$$D_{n1} = D_{n2} \text{ for } \rho_s = 0$$

5.7.2 Conductor - Free Space Interface :

We know that inside the conductor $\bar{D}$ and $\bar{E}$ are both zero. Consider medium 2 is a conductor, then under static conditions,

$$D_2 = E_2 = 0$$

$$\therefore E_{t2} = E_{t1} = 0 \quad \dots \text{from Equation (5.7.1)} \dots (5.7.5)$$

i.e. no tangential field present in medium 1 and medium 2. Now since $D_2 = 0$, Equation (5.7.3) reduces to,

$$D_{n1} = \rho_s \text{ or } E_{n1} = \frac{\rho_s}{\epsilon_0} \quad \dots (5.7.6)$$

The angle α_1 is obtained as, $\alpha_1 = \tan^{-1} \frac{E_{t1}}{E_{n1}} = \tan^{-1} \left(\frac{0}{E_{n1}} \right) \tan^{-1} (0) = 0$

It follows that a static electric field ($\bar{E}$) or flux density ($\bar{D}$) at a dielectric-free space boundary is always perpendicular to the conductor surface. The boundary conditions we developed for conductor - free space boundary are valid for the conductor - dielectric boundary if we replace ϵ_0 by ϵ .

For conductor dielectric interface.

$$D_t = E_t = 0 ; \quad D_n = \epsilon E_n = \rho_s$$

...(5.7.7)

5.7.3 Solved Examples on Electrostatic Boundary Conditions :

Important Formulae

For dielectric-dielectric interface

$$\vec{E}_{t1} = \vec{E}_{t2} \quad D_{n1} - D_{n2} = \rho_s$$

$$\text{or } D_{n1} = D_{n2} \text{ for } \rho_s = 0$$

For conductor-dielectric interface

$$D_t = E_t = 0; \quad D_n = \epsilon E_n = \rho_s$$

Example 5.7.1 : Given that $\vec{E}_1 = 2\vec{a}_x - 3\vec{a}_y + 5\vec{a}_z$ (V / m) at the charge free dielectric interface of shown in Fig. Ex. 5.7.1. Find D_2 and angles θ_1 and θ_2 .

MU - May 2001

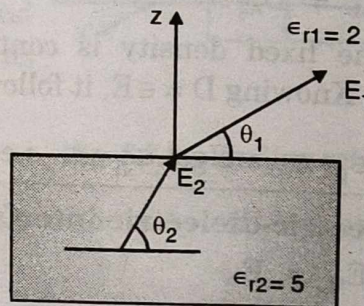


Fig. : Figure for Example 5.7.1

Solution : From the Fig. Ex. 5.7.1 it is clear that z-axis is normal to the boundary. Therefore,

$$\vec{E}_{n1} = 5\vec{a}_z \text{ (V / m)}$$

$$\vec{E}_{t1} = 2\vec{a}_x - 3\vec{a}_y \text{ (V / m)}$$

And

To find $\vec{D}_2$, we use

$$\vec{D}_2 = \vec{D}_{n2} + \vec{D}_{t2}$$

i.e. knowledge of D_{n2} and D_{t2} is required to determine $\vec{D}_2$.

To determine $\vec{D}_{t2}$:

From boundary conditions we have,

$$\vec{E}_{t2} = \vec{E}_{t1} = 2\vec{a}_x - 3\vec{a}_y$$

$$\therefore \vec{D}_{t2} = \epsilon_2 \vec{E}_{t2} = 5\epsilon_0 (2\vec{a}_x - 3\vec{a}_y) = 10\epsilon_0 \vec{a}_x - 15\epsilon_0 \vec{a}_y \text{ (C / m}^2\text{)}$$

To determine $\vec{D}_{n2}$:

We know the boundary condition at the charge free interface as,

$$\vec{D}_{n2} = \vec{D}_{n1} = \epsilon_1 \vec{E}_{n1} = 2\epsilon_0 (5\vec{a}_z) = 10\epsilon_0 \vec{a}_z$$

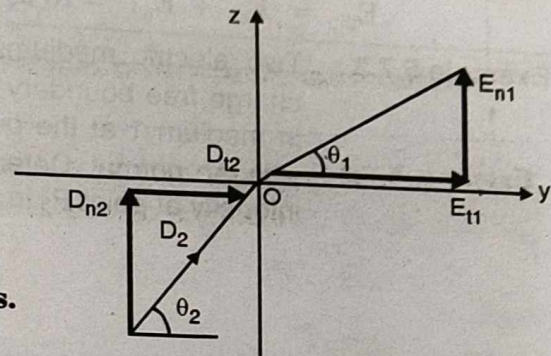
Using $\bar{D}_{n2}$ and $\bar{D}_{t2}$,

$$\begin{aligned}\bar{D}_2 &= 10 \epsilon_0 \bar{a}_x - 15 \epsilon_0 \bar{a}_y + 10 \epsilon_0 \bar{a}_z \text{ (C / m}^2\text{)} \\ &= 88.42 \bar{a}_x - 132.63 \bar{a}_y + 88.42 \bar{a}_z \text{ (pC / m}^2\text{)}\end{aligned}$$

Angles θ_1 and θ_2 is determined from the Fig. Ex. 5.7.1(a) as

$$\begin{aligned}\theta_1 &= \tan^{-1} \frac{E_{n1}}{E_{t1}} \\ &= \tan^{-1} \frac{5}{\sqrt{2^2 + (-3)^2}} = 54.2^\circ \quad \dots \text{Ans.}\end{aligned}$$

$$\begin{aligned}\text{and } \theta_2 &= \tan^{-1} \frac{D_{n2}}{D_{t2}} \\ &= \tan^{-1} \frac{10 \epsilon_0}{\epsilon_0 \sqrt{10^2 + (-15)^2}} = 29.0^\circ \quad \dots \text{Ans.}\end{aligned}$$



...Ans.

Fig. : Illustrating Ex. 5.7.1(a)

In this problem note the angles θ_1 and θ_2 which are not measured with normal but are measured with boundary surface, therefore, instead of $\theta_1 = \tan^{-1} (E_{t1} / E_{n1})$, in the previous problem it changes to $\tan^{-1} (E_{n1} / E_{t1})$. Similarly for θ_2 .

Example 5.7.2 : The plane $z = 0$ makes the boundary between the free space and dielectric medium with a dielectric constant 40. The electric field next to the interface in free space is $\bar{E} = 13 \bar{a}_x + 40 \bar{a}_y + 50 \bar{a}_z$ (V / m). Determine the electric field on the other side of the interface.

MU - Dec. 2004

Solution : From the given description the Fig. Ex. 5.7.2 can be drawn as shown.

$\bar{E}$ in free space is given, let it be $\bar{E}_1$. Since $z = 0$ forms the interface, the axis normal to interface is z -axis. Now $\bar{E}_1$ can be resolved into two parts,

$$\bar{E}_{n1} = 50 \bar{a}_z \text{ (V / m)}$$

$$\bar{E}_{t1} = 13 \bar{a}_x + 40 \bar{a}_y \text{ (V / m)}$$

To get $\bar{E}_2$ we required $\bar{E}_{n2}$ and $\bar{E}_{t2}$. Using boundary condition for tangential components,

$$\bar{E}_{t2} = \bar{E}_{t1}$$

$$\therefore \bar{E}_{t2} = 13 \bar{a}_x + 40 \bar{a}_y \text{ (V / m)}$$

To find $\bar{E}_{n2}$ we use boundary condition for normal components,

$$\text{i.e. } \bar{D}_{n2} = \bar{D}_{n1} = \epsilon_1 \bar{E}_{n1} = \epsilon_0 (50 \bar{a}_z)$$

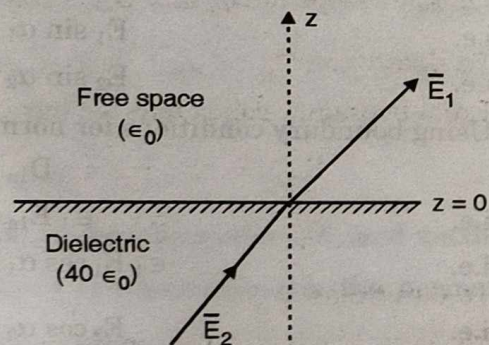


Fig. : Illustrating Ex. 5.7.2

To find $\bar{E}_{n2}$,

$$\bar{E}_{n2} = \frac{\bar{D}_{n2}}{\epsilon_2} = \frac{50 \epsilon_0 \bar{a}_z}{40 \epsilon_0} = 1.25 \bar{a}_z \text{ (V/m)}$$

The electric field in medium 2 is,

$$\bar{E}_2 = \bar{E}_{t2} + \bar{E}_{n2} = 13 \bar{a}_x + 40 \bar{a}_y + 1.25 \bar{a}_z \text{ (V/m)}$$

Example 5.7.3 : Two electric mediums with permittivities ϵ_1 and ϵ_2 are separated by a charge free boundary as shown in Fig. Ex. 5.7.3. The electric field intensity in medium 1 at the point P_1 has a magnitude E_1 and makes an angle α_1 with the normal. Determine the magnitude and direction of the electric field intensity at point P_2 in medium 2.

MU - May 2004

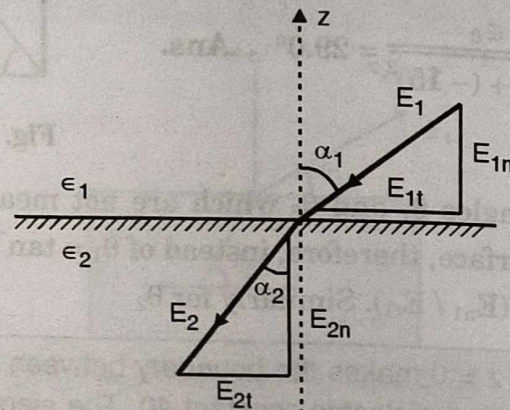


Fig. : Illustrating Ex. 5.7.3

Solution : In the problem electric field intensity E_1 is given with angle α_1 made with normal. It can be resolved into two parts E_{1t} and E_{1n} .

$$E_{1t} = E_1 \sin \alpha_1 \text{ and}$$

$$E_{1n} = E_1 \cos \alpha_1$$

Using boundary conditions for tangential components

$$E_{1t} = E_{2t}$$

i.e.

$$E_1 \sin \alpha_1 = E_2 \sin \alpha_2$$

i.e.

$$E_2 \sin \alpha_2 = E_1 \sin \alpha_1 \quad \dots(1)$$

Using boundary conditions for normal components

$$D_{1n} = D_{2n}$$

i.e.

$$\epsilon_1 E_{1n} = \epsilon_2 E_{2n}$$

i.e.

$$\epsilon_1 E_1 \cos \alpha_1 = \epsilon_2 E_2 \cos \alpha_2$$

i.e.

$$E_2 \cos \alpha_2 = \frac{\epsilon_1}{\epsilon_2} E_1 \cos \alpha_1 \quad \dots(2)$$

Squaring and adding Equations (1) and (2)

$$E_2^2 \sin^2 \alpha_2 + E_2^2 \cos^2 \alpha_2 = E_1^2 \sin^2 \alpha_1 + \left(\frac{\epsilon_1}{\epsilon_2}\right)^2 E_1^2 \cos^2 \alpha_1$$

$$E_2^2 = E_1^2 \left[\sin^2 \alpha_1 + \left(\frac{\epsilon_1}{\epsilon_2}\right)^2 \cos^2 \alpha_1 \right]$$

i.e.

$$E_2 = E_1 \sqrt{\sin^2 \alpha_1 + \left(\frac{\epsilon_1}{\epsilon_2}\right)^2 \cos^2 \alpha_1}$$

or

Dividing Equation (1) by Equation (2) we get,

$$\frac{\sin \alpha_2}{\cos \alpha_2} = \frac{\epsilon_2}{\epsilon_1} \frac{\sin \alpha_1}{\cos \alpha_1}$$

$$\tan \alpha_2 = \frac{\epsilon_2}{\epsilon_1} \tan \alpha_1$$

...Ans.

Example 5.7.4 : Region $z < 0$ be composed of uniform dielectric material for which $\epsilon_r = 3.2$ while region $z > 0$ is characterized by $\epsilon_r = 2$.

Let $\bar{D}_1 = -30 \bar{a}_x + 50 \bar{a}_y + 70 \bar{a}_z$ nC/m². Find (i) D_{N1} (ii) D_{t1}

MU - Dec. 2002

Solution : You should always remember that boundary conditions are used to find field vector in one medium when field vector in other is given. In the problem the vector $\bar{D}_1$ is given and it is asked to find D_{t1} and D_{n1} , which does not required boundary conditions at all.

The tangential and normal components of D_1 can be determined by finding the axis which is perpendicular to the boundary. Since, two regions are formed by $z < 0$ and $z > 0$, the z -axis is normal to the boundary. The components of D_1 in z -direction is normal component while remaining part of D_1 is tangential component.

Hence,

$$\bar{D}_{t1} = 70 \bar{a}_z \quad ; \quad \therefore D_{t1} = 70 \text{ nC/m}^2 \quad \text{...Ans.}$$

$$\bar{D}_{n1} = -30 \bar{a}_x + 50 \bar{a}_y \quad ; \quad \therefore D_{n1} = \sqrt{(-30)^2 + 50^2} = 58.31 \text{ nC/m}^2 \quad \text{...Ans.}$$

Example 5.7.5 : The region $x < 0$ contains dielectric medium for which $\epsilon_{r1} = 4$, while the region $x > 0$ is characterized by $\epsilon_{r2} = 2$, if $\bar{E}_1 = 50 \bar{a}_x - 30 \bar{a}_y + 60 \bar{a}_z$ (kV/m).

Find : (i) $\bar{E}_{N1}$; (ii) θ_1 ; (iii) E_{t1} ; (iv) $\bar{E}_2$; (v) θ_2 .

Solution : From the data given it is clear that the boundary surface is at $x = 0$, and x -axis is normal to the interface. Therefore component of $\bar{E}_1$ in the x -direction is the normal component while remaining expression of $\bar{E}_1$ is tangential component.

Thus,

$$\bar{E}_{N1} = 50 \bar{a}_x \text{ (kV/m)}$$

and $\bar{E}_{t1} = -30 \bar{a}_y + 60 \bar{a}_z \text{ (kV/m)}$

(i) To find $\bar{E}_{N1}$:

As mentioned above

$$\bar{E}_{N1} = 50 \bar{a}_x \text{ (kV/m)}$$

...Ans.

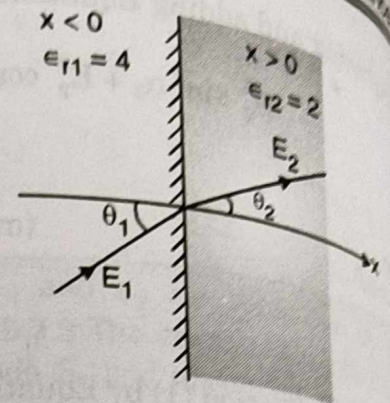


Fig. : Illustrating Ex. 5.7.5

(ii) To find θ_1 :

As it is note mentioned, whether θ_1 is measured with normal or with boundary, let us measure it with normal.

$$\therefore \theta_1 = \tan^{-1} \left(\frac{E_{t1}}{E_{N1}} \right) = \tan^{-1} \left[\frac{\sqrt{(-30)^2 + (60)^2}}{50} \right] = 53.30^\circ$$

...Ans.

(iii) To find E_{t1} :

Since $\bar{E}_{t1} = -30 \bar{a}_y + 60 \bar{a}_z \text{ (kV/m)}$

$$E_{t1} = \sqrt{(-30)^2 + (60)^2} = 67.1 \text{ (kV/m)}$$

...Ans.

(iv) To find $\bar{E}_2$:

$$\bar{E}_2 = \bar{E}_{t2} + \bar{E}_{N2}$$

We know that

$$\bar{E}_{t2} = \bar{E}_{t1} = -30 \bar{a}_y + 60 \bar{a}_z \text{ (kV/m)}$$

and $\bar{E}_{N2}$ is determined by using $\bar{D}_{N2}$.

Now

$$\bar{D}_{N2} = \bar{D}_{N1} = \epsilon_1 (E_{N1}) = 4 \epsilon_0 (50 \bar{a}_x)$$

$$\therefore \bar{E}_{N2} = \frac{\bar{D}_{N2}}{\epsilon_2} = \frac{4 \epsilon_0 (50 \bar{a}_x)}{2 \epsilon_0} = 100 \bar{a}_x \text{ (kV/m)}$$

Thus,

$$\bar{E}_2 = 100 \bar{a}_x - 30 \bar{a}_y + 60 \bar{a}_z \text{ (kV/m)}$$

...Ans.

(v) To find θ_2 :

The angle θ_1 can be determined either by determining E_2 and then considering its tangential and normal component or by using the formula.

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

$$\tan \theta_2 = \left(\frac{2 \epsilon_0}{4 \epsilon_0} \right) \tan (53.30)^\circ = 0.67$$

$$\theta_2 = \tan^{-1} 0.67 = 33.85^\circ$$

$$\theta_2 = \tan^{-1} \frac{E_{t2}}{E_{n2}} = \tan^{-1} \left(\frac{\sqrt{(-30)^2 + (60)^2}}{100} \right) = 33.85^\circ$$

...Ans.

Using E_2 ;

Example 5.7.6 : Prove the boundary value conditions for the electrostatic field.

A boundary exists at $z = 0$ between two dielectrics $\epsilon_{r1} = 2.5$ in region $z < 0$ and $\epsilon_{r2} = 4$ in the region $z > 0$. The field in the region of ϵ_{r1} is

$$\bar{E}_1 = -30 \bar{a}_x + 50 \bar{a}_y + 70 \bar{a}_z \text{ (V/m)}. \text{ Find}$$

- normal component of $\bar{E}_1$
- tangential component of $\bar{E}_1$
- angle θ_1 between $\bar{E}_1$ and normal to the surface (given $\alpha_1 \leq 90^\circ$)
- normal component of $\bar{D}_2$
- tangential component of $\bar{D}_2$
- angle θ_2 between $\bar{D}_2$ and normal to the surface

Solution :

- (i) **To find normal component of $\bar{E}_1$:** The interface of two media is at $z = 0$, so z -axis is normal to the interface. Any component in z -direction is the normal component.

$$\bar{E}_{n1} = 70 \bar{a}_z \text{ (V/m)} \quad \dots \text{Ans.}$$

- (ii) **To find tangential component of $\bar{E}_1$:**

The electric field $\bar{E}_1$ can be expressed as

$$\bar{E}_1 = \bar{E}_{t1} + \bar{E}_{n1}$$

so,

$$\bar{E}_{t1} = \bar{E}_1 - \bar{E}_{n1} = -30 \bar{a}_x + 50 \bar{a}_y \text{ (V/m)}$$

- (iii) **To find angle between $\bar{E}_1$ and normal :**

From Fig. Ex. 5.7.6,

$$\tan \theta_1 = \frac{|\bar{E}_{t1}|}{|\bar{E}_{n1}|} = \frac{\sqrt{30^2 + 50^2}}{70} = 0.8329$$

$$\theta_1 = \tan^{-1}(0.8329) = 39.79^\circ$$

...Ans.

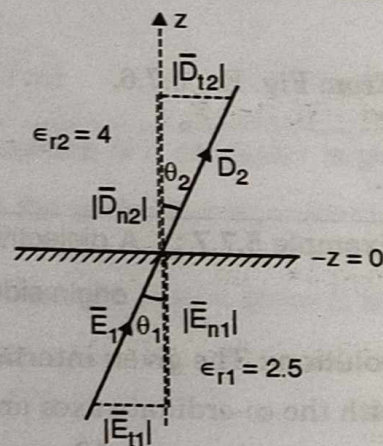


Fig. Illustrating Ex. 5.7.6

...Ans.

(iv) To find normal component of $\bar{D}_2$:

We have, $\bar{E}_{n1} = 70 \bar{a}_z \rightarrow \bar{D}_{n1} = \epsilon_1 \bar{E}_{n1} = 2.5 \epsilon_0 (70 \bar{a}_z)$

$$\bar{D}_{n1} = 175 \epsilon_0 \bar{a}_z \text{ (C/m}^2\text{)}$$

Using boundary condition

$$\bar{D}_{n2} = \bar{D}_{n1} = 175 \epsilon_0 \bar{a}_z \text{ (C/m}^2\text{)}$$

...Ans.

(v) To find tangential component of $\bar{D}_2$:

We have $\bar{E}_{t1} = -30 \bar{a}_x + 50 \bar{a}_y \text{ (V/m)}$

From boundary condition $\bar{E}_{t2} = \bar{E}_{t1} = -30 \bar{a}_x + 50 \bar{a}_y \text{ (V/m)}$.

The flux density is, $\bar{D}_{t2} = \epsilon_2 \bar{E}_{t2} = 4 \epsilon_0 (-30 \bar{a}_x + 50 \bar{a}_y)$
 $= -120 \epsilon_0 \bar{a}_x + 200 \epsilon_0 \bar{a}_y \text{ (C/m}^2\text{)}$

...Ans.

Adding normal and tangential component of $\bar{D}_2$ we get

$$\bar{D}_2 = \bar{D}_{t2} + \bar{D}_{n2} = -120 \epsilon_0 \bar{a}_x + 200 \epsilon_0 \bar{a}_y + 175 \epsilon_0 \bar{a}_z \text{ (C/m}^2\text{)}$$

...Ans.

(vi) To find angle θ_2 :

From Fig. Ex. 5.7.6, $\tan \theta_2 = \frac{|\bar{D}_{t2}|}{|\bar{D}_{n2}|} = \frac{\sqrt{(120 \epsilon_0)^2 + (200 \epsilon_0)^2}}{170 \epsilon_0} = 1.371$

$\therefore \theta_2 = \tan^{-1}(1.371) = 53.91^\circ$...Ans.

Example 5.7.7 : A dielectric-free space interface has the equation $3x + 2y + z = 12 \text{ m}$. The origin side of the interface has $\epsilon_{r1} = 3$ and $\bar{E}_1 = 2 \bar{a}_x + 5 \bar{a}_z \text{ (V/m)}$. Find $\bar{E}_2$.

Solution : The given interface equation represents a plane. The intersections of this plane with the co-ordinate axes are,

$$x = \frac{12}{3} = 4 \text{ m ; } y = \frac{12}{2} = 6 \text{ m ; } z = 12 = 12 \text{ m}$$

The interface is as shown in Fig. Ex. 5.7.7. The equation of normal to the plane $Ax + By + Cz = D$ is

$$\bar{a}_n = \frac{A \bar{a}_x + B \bar{a}_y + C \bar{a}_z}{\sqrt{A^2 + B^2 + C^2}}$$

Then for the given plane, $\bar{a}_n = \frac{3 \bar{a}_x + 2 \bar{a}_y + \bar{a}_z}{\sqrt{14}}$

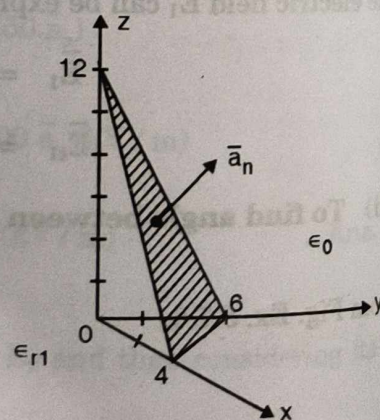


Fig. : Illustrating Ex. 5.7.7

Now, we should find normal and tangential component of given $\vec{E}_1$. The normal component is obtained by taking projection of $\vec{E}_1$ in the $\vec{a}_n$ direction. That is

$$E_{n1} = \vec{E}_1 \cdot \vec{a}_n = (2\vec{a}_x + 5\vec{a}_z) \cdot \left(\frac{3\vec{a}_x + 2\vec{a}_y + \vec{a}_z}{\sqrt{14}} \right) = \frac{11}{\sqrt{14}}$$

$$\vec{E}_{n1} = E_{n1} \vec{a}_n = \frac{11}{\sqrt{14}} \left(\frac{3\vec{a}_x + 2\vec{a}_y + \vec{a}_z}{\sqrt{14}} \right) = 2.36\vec{a}_x + 1.57\vec{a}_y + 0.79\vec{a}_z$$

The tangential component is, $\vec{E}_{t1} = \vec{E}_1 - \vec{E}_{n1} = -0.36\vec{a}_x - 1.57\vec{a}_y + 4.21\vec{a}_z$.

From boundary conditions, $\vec{E}_{t2} = \vec{E}_{t1} = -0.36\vec{a}_x - 1.57\vec{a}_y + 4.21\vec{a}_z$.

In order to obtain $\vec{E}_{n2}$, first we should find D_{n2} .

$$\vec{D}_{n1} = \epsilon_0 \epsilon_{r1} = 3\epsilon_0 (2.36\vec{a}_x + 1.57\vec{a}_y + 0.79\vec{a}_z) = \epsilon_0 (7.08\vec{a}_x + 4.71\vec{a}_y + 2.37\vec{a}_z)$$

From boundary conditions $\vec{D}_{n2} = \vec{D}_{n1}$.

$$\therefore \vec{E}_{n2} = \frac{1}{\epsilon_0} \vec{D}_{n2} = \frac{1}{\epsilon_0} \times \vec{D}_{n1} = \frac{1}{\epsilon_0} \times \epsilon_0 (7.08\vec{a}_x + 4.71\vec{a}_y + 2.37\vec{a}_z) = 7.08\vec{a}_x + 4.71\vec{a}_y + 2.37\vec{a}_z$$

The required electric field $\vec{E}_2$ is obtained as,

$$\vec{E}_2 = \vec{E}_{t2} + \vec{E}_{n2} = 6.72\vec{a}_x + 3.14\vec{a}_y + 6.58\vec{a}_z \text{ (V/m)} \quad \dots \text{Ans.}$$

Example 5.7.8 : The electric field intensity at a point on the surface of a conductor is given by, $\vec{E} = 0.2\vec{a}_x - 0.3\vec{a}_y - 0.2\vec{a}_z$ (V/m). Find the surface charge density at that point.

Solution : Since for a conductor tangential component of E is zero, it means given E must be a normal component.

$$\vec{E} = \vec{E}_n = 0.2\vec{a}_x - 0.3\vec{a}_y - 0.2\vec{a}_z \text{ (V/m)}$$

The flux density (D) must also be normal,

$$\begin{aligned} \vec{D}_n &= \epsilon_0 \vec{E}_n = (0.2\vec{a}_x - 0.3\vec{a}_y - 0.2\vec{a}_z) \times \frac{10^{-9}}{36\pi} \\ &= 1.77\vec{a}_x - 2.66\vec{a}_y - 1.77\vec{a}_z \text{ (pC/m}^2\text{)} \end{aligned}$$

From the boundary conditions the charge on the surface of a conductor is,

$$\rho_s = D_n = \sqrt{1.77^2 + 2.66^2 + 1.77^2} = 3.65 \text{ (pC/m}^2\text{)} \quad \dots \text{Ans.}$$

Example 5.7.9 : A potential field is given as

$$V = 100e^{-5x} \sin 3y \cos 4z \text{ (volts).}$$

Let point P (0.1, $\pi/12$, $\pi/24$) be located at conductor free space boundary. At

point P find the magnitude of (i) V (ii) $\vec{E}$ (iii) E_N (iv) E_t (v) ρ_s .

**Solution :****(i) To find V :**

Since the expression for V is in cartesian system, we should convert given point in cartesian.

$$x = r \sin \theta \cos \phi = 0.0256$$

$$y = r \sin \theta \sin \phi = 0.0034$$

$$z = r \cos \theta = 0.097$$

so $P(0.1, \pi/12, \pi/24) = P(0.0256, 0.0034, 0.097)$

$$V \text{ at } P = 100e^{-5(0.0256)} \sin 3(0.0034) \cos 4(0.097) = 0.0156 \text{ volts.}$$

(ii) To find $\vec{E}$:

$$\vec{E} = -\nabla V = -\left[\frac{\partial V}{\partial x} \vec{a}_x + \frac{\partial V}{\partial y} \vec{a}_y + \frac{\partial V}{\partial z} \vec{a}_z\right]$$

$$= -[-500 e^{-5x} \sin 3y \cos 4z \vec{a}_x + 300 e^{-5x} \cos 3y \cos 4z - 400 e^{-5x} \sin 3y \sin 4z]$$

Putting the values of x, y, z we get

$$\vec{E} \text{ at } P = +0.078 \vec{a}_x - 263.88 \vec{a}_y + 4.84 \times 10^{-6} \vec{a}_z \text{ (V/m)}$$

(iii) To find E_n :

At the conductor free space interface $\vec{E}$ or $\vec{D}$ both are normal to the conductor surface. So $\vec{E}$ in previous part is nothing but $\vec{E}_n$. The magnitude of it is

$$E_n = \sqrt{(0.078)^2 + (263.88)^2 + (4.84 \times 10^{-6})^2} = 263.88 \text{ (V/m)}$$

(iv) To find E_t :

From the boundary conditions the tangential component of $\vec{E}$ must be zero.

$$E_t = 0$$

(v) To find ρ_s :

From the boundary conditions

$$\rho_s = D_n = \epsilon_0 E_n = 8.854 \times 10^{-12} (263.88) = 2.34 \text{ (nC/m}^2\text{)}$$

Example 5.7.10 : Given the potential

$V = 10(x^2 + xy)$ and a point P (2, -1, 3) on a conductor to free space boundary. Find V, $\vec{E}$, $\vec{D}$ at a point P.

Solution :

To find V : V at P (2, -1, 3) = $10(2^2 - 1 \times 2) = 20$ (Volts)

To find $\vec{E}$:

$$\vec{E} = -\nabla V = -\left(\frac{\partial V}{\partial x} \vec{a}_x + \frac{\partial V}{\partial y} \vec{a}_y + \frac{\partial V}{\partial z} \vec{a}_z\right)$$

$$= -[10(2x + y) \vec{a}_x + 10(x) \vec{a}_y + 0]$$

$$\vec{E} \text{ at } P = -[10(4 - 1) \vec{a}_x + 20 \vec{a}_y] = -30 \vec{a}_x - 20 \vec{a}_y \text{ (V/m)}$$

To find $\vec{D}$:

$$\vec{D} = \epsilon_0 \vec{E} = 8.854 \times 10^{-12} (-30 \vec{a}_x - 20 \vec{a}_y)$$

$$= -0.266 \vec{a}_x - 0.177 \vec{a}_y \text{ (nC/m}^2\text{)}$$